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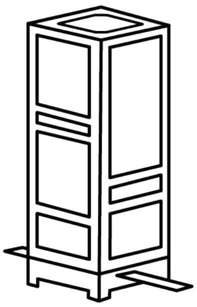
The Satellite Learning Laboratory is a fully functional satellite model designed as a hands on teaching aid and development environment for the classroom and laboratory.



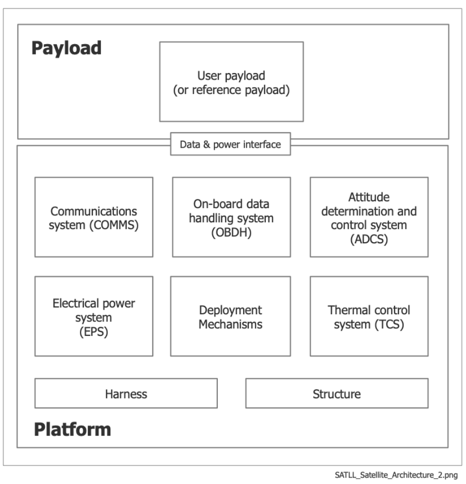
SATELLITE LEARNING LABORATORY

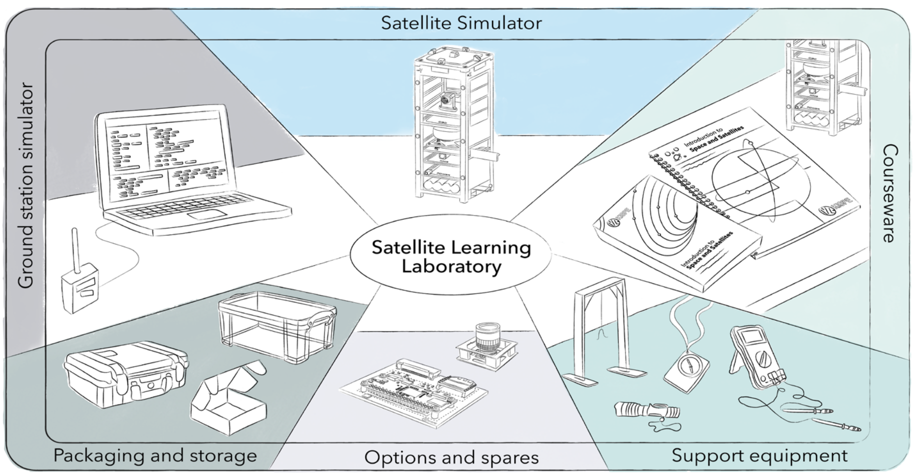
The Satellite Learning Laboratory can be used to support training in:

- Spacecraft systems
- Assembly integration and test
- Each sub systems can be tested individually before integration into complete unit
- Systems engineering, verification and validation
- Operations

The Satellite Learning Laboratory demonstrates the following satellite subsystems:

- Imaging Payload
- Electrical Power (EPS)
- On-Board Data Handling (OBDH)
- Communications (Comms)
- Deployment mechanisms
- Attitude Determination and Control (ADCS)
- Thermal
- Structure
- Launch integration and deployment





SATELLITE LEARNING LABORATORY SPECIFICATIONS

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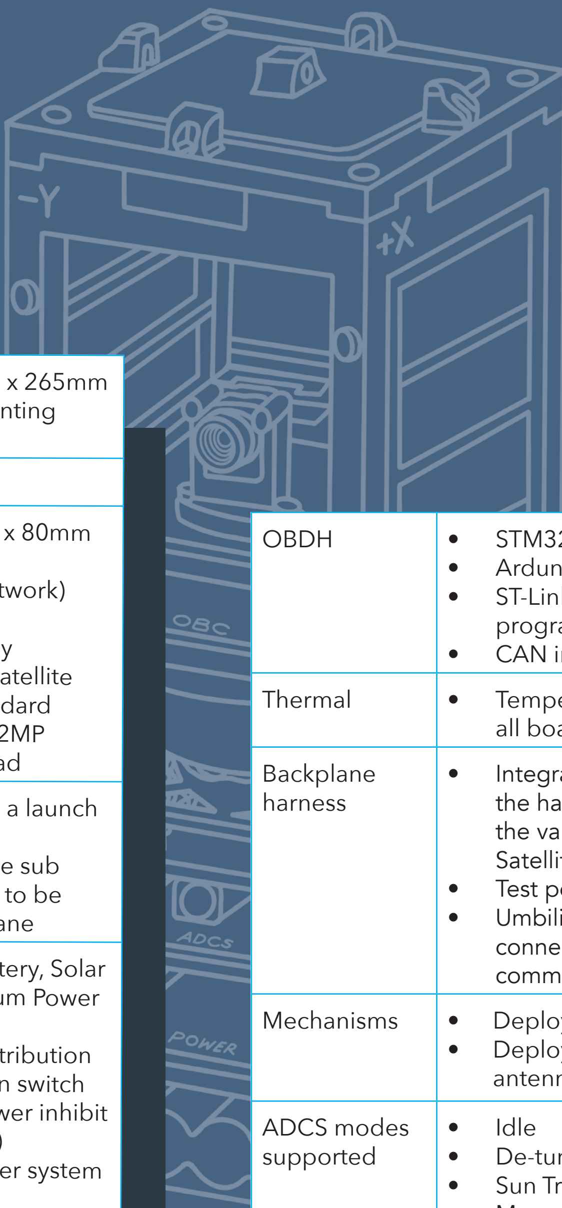

Size	• Approximately 100 x 100 x 265mm structure (excluding mounting points)
Mass	• <3 kg
Payload	Modular payload bay (80mm x 80mm x 80mm) • CAN (Controller Area Network) platform interface • 5V switched power supply • Standard payload - The Satellite Learning Laboratory Standard Kit includes the OV2620 2MP ArduCam imaging payload
Structure	• 3D printed structure with a launch vehicle dispenser • Mechanically supports the sub systems and allows them to be connected to the backplane
EPS	• 7.4V 5000mAh Li-Ion battery, Solar Array with MPPT (Maximum Power Point Tracking) charging • Power regulation and distribution • Launch vehicle separation switch • Remove before flight power inhibit (classroom power switch) • Launch deployment power system sequencing • Power regulation (5V and 3.3V) • Power system telemetry (Voltage, Current, Status) • Umbilical connection for ground-based power supply during testing.
ADCS	• STM32 based ADCS computer • Deployable Magnetometer, • 3 axis Magnetorquers, • Reaction wheel and motor control board, • Sun sensors and • IMU (incorporating a gyro)
Communications	• 434MHz (Asia), 868 MHz (Europe) or 915 MHz (US) Ultra High Frequency (UHF) communications half duplex transceiver. • UHF band antenna • RP2040 based CAN node for communication module telemetry and telecommand. • Pre-allocated unit operating frequency, paired with corresponding ground station simulator. • Operating frequency and power level modifiable by telecommand. • 200kbps default data rate.

OBDH	• STM32 microprocessor OBC • Arduno IDE for programming • ST-Link adapter for advanced programming • CAN interface to sub systems
Thermal	• Temperature sensors included on all boards
Backplane harness	• Integrated backplane providing the harness connections between the various subsystems within the Satellite Simulator. • Test points • Umbilical connector for external connection to power and the CAN communications bus
Mechanisms	• Deployable magnetometer • Deployable "Tape Measure" UHF antenna
ADCS modes supported	• Idle • De-tumbling (1 axis) • Sun Tracking (1 axis) • Manual
Manuals/ Training Material	• Quick Start Guide • User manual • Workbook
Platform Software	Platform: • Beginner/ Intermediate: Arduino • Advanced : RTOS (Option)
Ground Station Simulator	• User Friendly SCOTTI ground station software and radio provided, includes: • Operator mode • Developer mode • Test Mode
Required Equipment* **Items can be supplied as part of the optionsl suspsension frame and basic experimentation kit*	• Support frame for hanging (for ADCS experiments) • Torch (for power experiments) • Compass • Multimeter • Battery charger • Windows computer, with USB port
Optional Extras	• Suspension Frame • Basic Experimentation kit • Subsystem Bench Test Module • Software Development Kit • Additional Payloads • Flatsat motherboard • Laptop with preloaded ground station software